Compensation for Nasal Place Assimilation at Three Anchors of an Articulatory Continuum

A statistical companion note after the human–model comparison of Pouw et al. (2024), with $\lambda = 0.75$ and $\lambda = 0.625$ production reproduction attempts and a crossing with the natural stimuli of the authors' released archive

Companion note to *Nasal Place Assimilation in wav2vec2: A Didactic Synthesis*

October 2026

In one paragraph. The demonstration page that accompanies this research program keeps three steps of the articulatory assimilation continuum—$\lambda = 0$ (canonical cluster), $\lambda = 0.5$ (the gestural midpoint), and $\lambda = 1.0$ (C1 target on the C2 place coordinates)—because they carry the essential structure of the phenomenon. Under the revised reference model (`covtl-lambda` v1.2.0), the continuum is target-level and cluster-scoped: λ rotates the polar target of the C1 nasal of a tautosyllabic cluster toward the target of C2, while C2, the context, the lips, the velum and the glottal stream stay bitwise canonical; $\lambda = 0$ is bitwise the canonical cluster synthesis. This note quantifies the three anchor conditions with the statistical apparatus of the human–model comparison in Pouw, de Heer Kloots, Alishahi, and Zuidema [10]: compensation rates with 95% Wilson score intervals, a grouped-bar presentation in the style of their Figure 1, rank-order (Spearman) associations tested against an exact null distribution, and a claim-by-claim mapping of their findings onto ours. The headline is different from every earlier edition of this note, and it is a *sign inversion* at the rate level: at $\lambda = 1$ the licensed pool compensates *less* than the unlicensed pool for both listeners (69.2% against 89.7% for the CTC-tuned one, 82.1% against 100.0% for the self-supervised one; -20.5 $[-37.3, -2.4]$ and -17.9 $[-32.7, -5.3]$, both Newcombe intervals excluding zero), because the late, letter-supervised layers read /m/ on the labial-C2 clusters whose context already produced that reading at $\lambda = 0$. At the phonetic layers (T2–T4) both listeners hold the underlying /n/ at every λ of every frame—there is no category boundary anywhere on the revised continuum—and the $\lambda = 0$ control is now itself informative: the tautosyllabic cluster canonical reads /m/-ish at the acoustic layer (48.7% pooled compensation for both listeners), a coarticulation property that the dotted citation forms of the earlier tiers do not have.

This revision: a reproduction attempt at $\lambda = 0.75$. The anchor comparison above splits at $\lambda = 1$. The published stimuli were produced by a human speaker, and natural assimilation is partial. This revision therefore retries the comparison with $\lambda = 0.75$ (a strong but incomplete target rotation) as the production degree, over six sentence frames—three licensed (*clean pan*, *mean boat*, *thin packet*), three unlicensed (*clean tan*, *clean can*, *mean goat*)—in the exact three-condition protocol of their Figure 1 (Section 5). The attempt reproduces the published *levels* on the high side: compensation stands at 79.5% and 100.0% in the viable condition and 89.7% and 97.4% in the unviable one, at or above every published cell (46–75% viable, 11–40% unviable). It does not reproduce the published *asymmetry*: the viable–unviable difference is -10.3 points (95% CI $[-26.5, +6.2]$) for the CTC-tuned listener and $+2.6$ $[-6.6, +13.2]$ for the self-supervised one, against $+35$ points in both published populations. At every production degree tested, the models compensate according to how the gesturally confined rotation moves their representation, not according to whether the context licenses the surface assimilation.

Second addition: a $\lambda = 0.625$ production. If the asymmetry is absent at $\lambda = 1$ and at $\lambda = 0.75$, the remaining degree-side hypothesis is that a licensing modulation needs a stronger residual cue to express itself. A further run at $\lambda = 0.625$—five eighths of the way along the target rotation, same six frames, same protocol—raises the viable level of the self-supervised listener to 100% (above the published model cell of 75%) and leaves the CTC-tuned viable cell at 79.5%, with unviable cells at 89.7% and 94.9%, still far above the published unviable band. The asymmetry remains absent: -7.7 points $[-23.7, +8.4]$ for the CTC-tuned listener, $+5.1$ $[-4.6, +16.9]$ for the self-supervised one. With three degrees tested, the

viable–unviable differences wander in sign around zero at the moderate degrees (Section 6); the only intervals that exclude zero anywhere in the note are the $\lambda = 1$ licensed dips—of *both* listeners—and they point in the direction *opposite* to the published asymmetry.

Third addition: crossing with the natural stimuli. Every result above lives on our own synthesis, and the obvious objection is that the null asymmetry might be an artifact of the synthetic domain. The original Darcy et al. [3] recordings behind Pouw et al.'s Figure 1 are not public, but the authors released their own natural re-recordings of the Gaskell and Marslen-Wilson materials (86 sentences, the `lexamb-stimuli.zip` linked from their repository). We ran both of our measures on that archive (Section 7). The transcription measure replicates the published behavioral level: 14.5% compensation on the viable natural stimuli against their digitized 15.8%, with the two recorded passes agreeing. The probe measure then delivers the crossing result: on the natural assimilated [m] frames the self-supervised listener shows the full deep-layer shift toward the underlying /n/ (0/20 verdicts at the CNN, 17/20 at T12), while the CTC-tuned listener shows none (0/20 at T12, at every window margin) —whereas on the synthetic licensed stimuli of the moderate degrees its representational compensation was the majority pattern (79.5% pooled at $\lambda = 0.75$). The five velar-nasal items that Pouw et al.'s measure cannot score are recovered by our {n, m, N} probes (9/10 verdicts /n/ at T12 for the self-supervised listener). What the released archive cannot provide is the unviable cell—it contains only viable-context recordings—so the asymmetry question stays open on the natural side.

Fourth revision (v2.1): the C1-confined morph. All λ-experiments documented in this note were re-run under a corrected morphing protocol. The previous blend spanned the whole interval over which the endpoint variants differ—frames 97–307 of 624 for the licensed pair—which includes an anticipatory island on the vowel and the release transition of the following consonant; the morph therefore transformed part of the context it was supposed to hold fixed. The corrected protocol confines the blend to the nasal gesture window (frames 232–272, 0.58–0.68 s, with 25 ms raised-cosine ramps; the covtl-lambda-patch v1.1.0 of the engine), leaving every frame outside it bitwise canonical at every λ and the following consonant strictly invariant (asserted per stimulus). The $\lambda = 0$ stimuli are byte-identical to the previous version (md5-checked), so every Control condition carries over unchanged. The correction leaves the headline conclusions intact—no viable–unviable asymmetry at any production degree—and removes one artifact: the weak positive asymmetry of the self-supervised listener at $\lambda = 0.625$ ($+10.3$ points) was carried by the morphed transitions and disappears confined (-2.6). Compensation levels rise modestly with the clean context (the viable cells gain 2 to 10 points, the self-supervised unviable cell at $\lambda = 0.625$ rises from 76.9% to 97.4%), and the $\lambda = 1$ confined hybrids acquire one additional underlying verdict each. All numbers quoted below are the confined ones; the superseded unconfined results remain in `work/legacy_unconfined/` with a machine-readable comparison (`work/correction_comparison.json`, `CORRECTIONS_v2.1.md`).

Fifth revision (v3.0): the target-level cluster model. The λ reference has moved again, and this time the change is semantic, not procedural: the upstream repository `covtl-lambda` (v1.2.0) redefines the assimilation as a property of the *target* the cluster planning pursues, not of the motor trajectory. The λ variation now rotates the polar target of the C1 nasal of a tautosyllabic cluster toward the target of C2,

$$\theta_{C1}(\lambda) = \theta_{C1} + \lambda \cdot \mathrm{wrap}(\theta_{C2} - \theta_{C1}), \qquad \lambda \in [0, 1],$$

with four run-time guarantees recorded per synthesis ($\lambda = 0$ bitwise canonical; θ equality at $\lambda = 1$; glottal block bitwise constant; context outside the C1-incident arcs bitwise canonical). The dotted endpoint spellings of every earlier tier (`klin.pan`/`klim.pan`) are rejected by the reference engine—a dot is a syllable split, a different gesture structure—and the cluster spellings (`klinpan` &c.) take their place. All λ experiments of this note were re-run on that reference as a single 54-item campaign (six frames $\times$ nine degrees, both encoders, one probe instrument), so every number below changes; the v2.1 confined-motor results are preserved in `work/legacy_motorblend/` with the machine-readable comparison `work/revision_comparison_v30.json`. Three consequences organize the new reading. First, the lips never close on the assimilated side of the continuum: the labial constriction remains the business

of C2, so the late-layer /m/ readings that appear at $\lambda = 1$ are driven by the cluster context, not by a surface labial closure—P4 (*thin packet*), which contains no /m/ segment anywhere, reads /m/ at T10 along the whole continuum. Second, the phonetic layers (T2–T4) read the underlying /n/ at every λ of every frame: the category boundary of the earlier editions (T4 $\lambda^* \approx 0.7$–0.9) no longer exists, because the motor blend that produced it moved the lips. Third, the $\lambda = 0$ control changes meaning: it is now the canonical *cluster* (bitwise identical to the canonical synthesis), whose coarticulated nasal reads /m/-ish at the acoustic layer—a property of natural clusters that the dotted citation forms lacked, and the reason the pooled control sits at 64.1%/61.5% instead of near the ceiling.

Sixth revision (v3.1): the full CC selector. The v3.0 campaign kept one restriction whose cost the wav2vec2 readings made explicit: the retargeted C1 carried *its own* selector, so the articulators that realize C2 and are absent from sel(C1)—the lips LD/LP of a labial C2—stayed on the vocalic background along the whole continuum (the "lip blocking": lips bitwise constant on every frame). This revision removes it: under λ the *whole CC selector* is taken into account, as in the base model, whose cluster machinery already drives the union sel(C1)∪sel(C2). Concretely each degree is synthesized twice through the engine's own chain—once with C1's selector (the v3.0 item, bitwise), once with the union selector—and the C2-only parameters are blended from their canonical background values toward their full-selector realization with weight λ. The jaw/tongue rotation is bitwise identical to v3.0 at every degree; the glottal block, C2 and the context stay bitwise canonical; $\lambda = 0$ reproduces the canonical cluster bitwise; and for the unlicensed frames, whose C2 selectors (alveolar, velar) are subsets of sel(/n/), the condition is a structural no-op verified bitwise—the NV continua, and every $\lambda = 0$ anchor, carry over from v3.0 unchanged. On the licensed frames the lips now close in proportion: LD at the C1 target point runs $0.61 \to 0.46 \to 0.16 \to 0.00$ cm at $\lambda = 0/0.5/0.75/1$ (full labial closure, the base model's own /p/ value). The perceptual structure survives the completed gesture and is sharpened around it: the phonetic layers still hold the underlying /n/ at every degree of every frame (no T4 boundary), the $\lambda = 0.75$ rates of the CTC-tuned listener do not move at all, and the $\lambda = 1$ licensed dip—far from being an artifact of the open lips—*strengthens* and becomes bilateral: -20.5 $[-37.3, -2.4]$ (CTC-tuned) and -17.9 $[-32.7, -5.3]$ (self-supervised). Most tellingly, P1 (*clean pan*), the frame whose lips now close, *loses* its late-layer labial reading ($P(\mathrm{m})$ at T10–T11, $\lambda = 1$: $.44/.53 \to .24/.09$), while P2/P4 keep theirs: the letter-supervised surface event tracks the cluster *context*, not the surface closure. The v3.0 results are preserved verbatim under `work/legacy_lipblocked/` with the machine-readable comparison `work/revision_comparison_v31.json`.

1 Purpose and scope

The λ-continuum experiment of the synthesis report is now defined on the tautosyllabic clusters of six sentence frames, `klinpan` ("clean pan"), `minbot` ("mean boat"), `Tinpakit` ("thin packet"), `klintan` ("clean tan"), `klinkan` ("clean can") and `mingot` ("mean goat"), with the articulatory engine of the covtl pipeline [2] and the λ effect of the reference repository `covtl-lambda` v1.2.0. In every frame the C1 of the cluster is /n/ at polar target $(1.15, 270°)$; λ rotates that target toward the C2 target along the shorter arc ($+140°$ labial for the licensed frames, $+75°$ velar for NV3/NV4, $+15°$ nearly place-neutral for NV1), re-running the engine's own syllabic trajectory builder per degree with the C1 node retargeted. At $\lambda = 0$ the synthesis is bitwise identical to the canonical cluster; at $\lambda = 1$ the C1 target has reached the C2 coordinates—the place-assimilated cluster C1'C2—while C2 itself, the glottal block and the velum never move, and the lips *join* the gesture: under this revision the whole CC selector (the union of the two consonant selectors, i.e. the base model's own cluster machinery) is taken into account, the C2-only lip parameters recruited with weight λ (four run-time guarantees plus three full-selector checks, verified per synthesis and held for all 54 items; for the unlicensed frames the union reduces to sel(/n/) and the continuum is bitwise the blocked-lip one). Each of the 54 stimuli (six frames $\times$ nine degrees of λ) is passed through both wav2vec2 listeners—the self-supervised `wav2vec2-base` and the CTC-tuned `wav2vec2-base-960h`—and the nasal probe trained on the original training base returns, for each of the 13 network levels (the convolutional front end CNN and transformer layers T1 through T12), the three-

way probability over {n, m, ŋ} for the consonantal closure, following the layer-wise probing tradition of Pasad et al. [9]. The probe instrument is unchanged from the stage-1 report (leave-one-sentence-out accuracies identical to it), so the three tiers of λ experiments—dotted unconfined, dotted confined, and the present cluster-target one—are measured by the same instrument and remain directly comparable.

The interactive demonstration built from these materials deliberately keeps only three steps—$\lambda = 0$, 0.5, and 1.0—after an earlier version with more variants was judged visually overloaded. Those are the three *conditions* whose statistics this note documents: the $\lambda = 0$ endpoint is the canonical cluster whose early-layer coarticulation reading this note documents, the $\lambda = 1$ endpoint is the target-assimilated cluster whose late-layer labial reading this note documents, and the midpoint is the condition that separates a graded, gestural reading of the continuum from a categorical one. Everything else in the kit—the synthesis report, the probing browser, the reproduction scripts—is untouched by this note, which should be read as a statistical marginal note to the report; the 56-item dotted base of the earlier tiers is retained unchanged as the stage-1 measurement tier and as the probe training base.

2 The comparison mapping

Pouw et al. compare a speech-recognition model with human listeners on compensation for place assimilation: the listener is said to *compensate* when, hearing an assimilated form such as `kleam` in "clea[m] pan", they recover the underlying final consonant of "clean" rather than accepting the surface labial [3–5]. Their comparison rests on four instruments: compensation rates per condition (their Figure 1, with 95% Wilson confidence intervals over $N = 48$ stimuli per condition), a viable-versus-unviable contrast in phonological context, a Spearman correlation between the model's probability for the underlying consonant and bigram frequency ($\rho(76) = .40$, $p < .001$), and an interpretability layer locating where in the network the surface interpretation dominates. We have no human listeners; what we have is something the published comparison lacks—two otherwise identical models that differ only in supervision (letter-supervised CTC versus purely self-supervised). The mapping in Table 1 is therefore structural, not literal, and its single most important convention is that the role of the two populations of the human–model comparison is played by our two model variants: the CTC-tuned model is the "letter-supervised listener", the self-supervised model the "acoustic listener".

Two rows of the mapping deserve a comment before the numbers. First, Pouw et al.'s unviable condition uses assimilations that English phonology does not license, exactly as our NV1 context does: the acoustics say [m] where the grammar would keep [n]. Second, their control condition—an unassimilated target in a neutral frame—is structurally our $\lambda = 0$ anchor, with one difference worth keeping in mind: our control is the zero point of the very continuum under test, so the three conditions are not three separate stimulus sets but three slices of one morph, which is what makes the rank statistics of Section 9 possible at all.

Construct	Pouw et al. (2024)	This note
Listeners	wav2vec2 versus 83 human participants (cross-modal priming; 6)	self-supervised `base` versus CTC-tuned `base-960h`; supervision plays the role of population
Compensation	underlying consonant transcribed despite assimilated acoustics	probe verdict /n/ (underlying) rather than /m/ (surface) or /ŋ/
Viable context	e.g. clea[m] pan (labial trigger)	licensed pool at λ: `klinpan`, `minbot`, `Tinpakit` before /p/, /b/, /p/
Unviable context	e.g. grea[k] match (no triggering segment)	unlicensed pool: `klintan`, `klinkan`, `mingot` before /t/, /k/, /g/
Control	clea[n] fork (no assimilation applied)	the $\lambda = 0$ anchors (canonical clusters of both pools)
Rate and interval	compensation rate, 95% Wilson CI, $N = 48$ stimuli/condition	compensation rate over the 13 layer probes of one stimulus, 95% Wilson CI (descriptive)
Association test	Spearman $\rho(76) = .40$ against bigram frequency	Spearman $\rho(\lambda, P(\mathrm{n}))$ per layer across the 9-step continuum, exact null over the 9! permutations
Underlying probability	$M = 0.14$ (biasing) versus $M = 0.17$ (neutral) context, no effect	deep-layer $P(\mathrm{n})$ at $\lambda = 1$: licensed versus unlicensed context

Table 1: Construct-by-construct mapping between the human–model comparison of Pouw et al. and the three-condition statistics of this note.

3 Measures

Compensation rate. For a given stimulus (one listener, one context, one λ anchor) the 13 probes each return a verdict, the argmax over {n, m, ŋ}. The compensation rate is the proportion of verdicts equal to /n/, the underlying nasal. Verdicts /ŋ/ count neither as compensation nor as surface acceptance; they are a third outcome that turns out to matter only in one cell of the design (the deep layers of the CTC-tuned model at licensed $\lambda = 1$). Because a proportion with small denominator is poorly summarized by a Wald interval, we use the Wilson score interval at 95%,

$$\frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \;\pm\; \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}, \qquad z = 1.96,\ n = 13, \tag{1}$$

the same interval Pouw et al. report in their Figure 1, where $\hat{p} = k/13$ and k is the number of /n/ verdicts. One caveat is essential and is repeated in Section 11: our 13 units are network levels of a single stimulus, not independent items, so the interval describes how the verdicts disperse across depth; it is not a sampling interval over stimuli as in the original, and no inferential claim should be built on it.

Continuum fidelity. For each layer we compute the Spearman correlation between λ (nine steps) and $P(\mathrm{n})$ across the continuum, the analog of Pouw et al.'s correlation of the underlying consonant's probability with bigram frequency. With nine continuum steps the asymptotic t approximation is unsafe, so p-values are exact, obtained by enumerating all $9! = 362{,}880$ permutations of the ranks; two-sided significance at $\alpha = .05$ requires $|\rho| \geq 0.70$. Under the revised model the informative sign is *positive* for the unlicensed (velar-C2) pools: rotating the C1 target away from the labial-coarticulated cluster canonical moves the murmur *toward* the trained /n/ centroid, so $P(\mathrm{n})$ rises with λ (the pooled unlicensed ρ reaches $+.90$ and $+.98$ at T4). Small $|\rho|$ therefore means the layer has lost fidelity to the continuum—which, at depth, is exactly where the interesting behavior lives.

Category boundary. For each layer we interpolate the value λ^* at which $P(\text{n})$ first falls below 0.5 (and, separately, below $P(\text{m})$) along the nine steps. This is the probe analog of the perceptual category boundary along an identification continuum; a boundary near 1.0 means the surface interpretation wins only for nearly complete assimilation.

Deep contrast and agreement. We summarize each listener by (a) the deep-layer contrast $P(\text{n})|_{\lambda=1} - P(\text{n})|_{\lambda=0}$ at T11 and T12, and (b) the layer-wise Spearman agreement between the two listeners' $P(\text{n})$ across the continuum, which asks where the two "populations" of this note tell the same story.

4 Compensation rates at the three conditions

Table 2 gives the compensation rates of the three anchor conditions, pooled over the three frames of each pool (39 probes per cell), and Figure 1 (panels a and b) presents them in the grouped-bar format of Pouw et al.'s Figure 1. Three facts organize the reading, and each of them is new to this revision. First, the *control* anchors are no longer near the ceiling: at $\lambda = 0$ the licensed pool stands at 48.7% for both listeners and the unlicensed pool at 79.5% and 74.4%. The reason is articulatory, not pathological: the canonical tautosyllabic cluster carries anticipatory coarticulation of the C1 nasal with the following consonant, and the acoustic layer (CNN, T1) reads it as /m/-ish on the labial-C2 frames (pooled $P(\text{m})$ at the CNN $= .51$–$.54$ licensed against $.42$–$.44$ unlicensed), with the "min" onsets (P2, NV4) adding their lexical /m/ drift at the same depths. The dotted citation forms of the earlier tiers—carefully produced, syllable-separated [n]—lacked exactly this property, which is why their controls sat at 77–100%. Second, at the *midpoint* $\lambda = 0.5$ the underlying nasal dominates everywhere: 79.5% and 100% licensed, 89.7% and 100% unlicensed. The gestural rotation has now moved the tongue body away from the coarticulated closure, and every listener's majority reading is the underlying one—at the gestural midpoint of the revised morph the representation is *more* canonical than at $\lambda = 0$, not less. Third, at $\lambda = 1.0$ the design splits along new lines: the unlicensed pools hold (89.7% and 100%), while the licensed pools *dip*—69.2% for the CTC-tuned listener (-20.5 points against the unlicensed pool, Newcombe interval $[-37.3, -2.4]$) and 82.1% for the self-supervised one (-17.9 points, interval $[-32.7, -5.3]$)—the two intervals in the anchor table that exclude zero, and the only ones anywhere in the note. The dip is the late-layer labial reading, and the full-selector condition pins down what it is *not*: not the surface closure. At T10–T11, $\lambda = 1$, the labial-C2 clusters whose context already produced the reading at $\lambda = 0$ keep it (P2/P4: $P(\text{m}) = .65$–$.95$), while P1—whose lips now actually close—*loses* it ($P(\text{m}) = .24$ and $.09$, against $.44$ and $.53$ with the lips blocked): a cluster-context effect, not a closure effect. The direction of the licensed–unlicensed contrast at $\lambda = 1$ is therefore *inverted* relative to the published asymmetry: it is the licensed pool that loses underlying verdicts, and it loses them to the letter-supervised surface reading, not to the gestural rotation.

In the framework of Table 1, the $\lambda = 1$ row is our viable-versus-unviable comparison, and its honest summary at the rate level is an *inverted* null: the licensed pool compensates less than the unlicensed pool for both listeners, significantly so for both of them (-20.5 points, Newcombe interval $[-37.3, -2.4]$, and -17.9 $[-32.7, -5.3]$). Against the published asymmetry of $+35$ points in both populations, this is not a weaker version of their effect but the opposite sign, carried by a specific mechanism: the letter-supervised late layers reading /m/ on labial-C2 clusters—exactly the layers, and exactly the context, where the published interpretability analysis locates the surface dominance. The published comparison asks whether compensation *survives* assimilation; the revised continuum shows the listeners never needing to compensate at the phonetic layers at all (there is no category boundary to cross), while the late layers of the supervised model actively entertain the assimilated reading that the gesturally confined rotation never fully realizes acoustically. The next two sections test whether the production degree modulates this structure.

Figure 2 sets the two comparisons side by side in the format of Pouw et al.'s own Figure 1: one bar per condition, grouped by listener, Wilson intervals. The left half is the published comparison, reproduced—bar heights digitized from their figure, intervals recomputed from the implied $k/48$: compensation is

Pool	λ	CTC-tuned 960h	self-sup. base
Licensed (P1, P2, P4)	0 (control)	48.7 [34.2, 63.6]	48.7 [34.2, 63.6]
	0.5	79.5 [64.1, 89.3]	100.0 [91.0, 100.0]
	1.0	**69.2** [53.6, 81.4]	**82.1** [67.3, 91.0]
Unlicensed (NV1, NV3, NV4)	0 (control)	79.5 [64.1, 89.3]	74.4 [59.0, 85.4]
	0.5	89.7 [76.0, 95.8]	100.0 [91.0, 100.0]
	1.0	89.7 [76.0, 95.8]	100.0 [91.0, 100.0]

Table 2: Compensation rates (%, verdict /n/) of the 13 layer probes at the three anchor conditions, pooled over the three frames of each pool (39 probes per cell), with 95% Wilson score intervals (Equation 1). Verdict profiles at $\lambda = 0$: the licensed pool loses CNN–T1 (coarticulation reading, all three frames) and, on P2 and P4, T9–T11 (lexical /m/ of the "min"-free onsets plus the cluster context); the unlicensed pool loses CNN–T1 on NV4 ("min" onset). At $\lambda = 1$ the licensed dip of both listeners is the T10–T11 labial reading, carried by P2/P4 (which already read /m/ there at $\lambda = 0$); P1, whose lips close under the full-selector condition, loses no verdict at all in the CTC-tuned listener (13/13 /n/) and only the CNN (to [N]) in the self-supervised one.

high in viable contexts (75% for the model, 46% for the humans of 3), lower but far from absent in unviable ones (about 40% and 11%), and at ceiling on the controls. That viable–unviable asymmetry is what anchors their claim of sensitivity to phonological context. The right half appends this study's two listeners, whose bars are assembled along the licensed continuum—Control ($\lambda = 0$, canonical cluster), the gestural midpoint $\lambda = 0.5$ (hatched, the condition the published design cannot express), Viable ($\lambda = 1$, target-assimilated cluster)—with the unlicensed $\lambda = 1$ pool one step outside the continuum bracket. Read in this common format, our result is not a miniature of theirs but a different shape: the control bars sit low (the coarticulated cluster canonical), the midpoint bars are the highest of each group, and the viable bar sits *below* the unlicensed one for the CTC-tuned listener—the inverted asymmetry of Table 2. The $\lambda = 0.5$ bars carry the clearest message of the revised design: at the gestural midpoint the underlying reading is the majority reading everywhere (79.5–100%), and the earlier editions' "collapse between $\lambda = 0.5$ and $\lambda = 1$" is gone, because the motor blend that produced it closed the lips abruptly, whereas the proportional recruitment of the full-selector rotation does not trigger it either. Where the published figure asks whether the output compensates, ours locates where along the morph the verdicts stop compensating; under the revised reference they stop only at the late supervised layers, only in the licensed pool, and only at the very end of the rotation (Section 8).

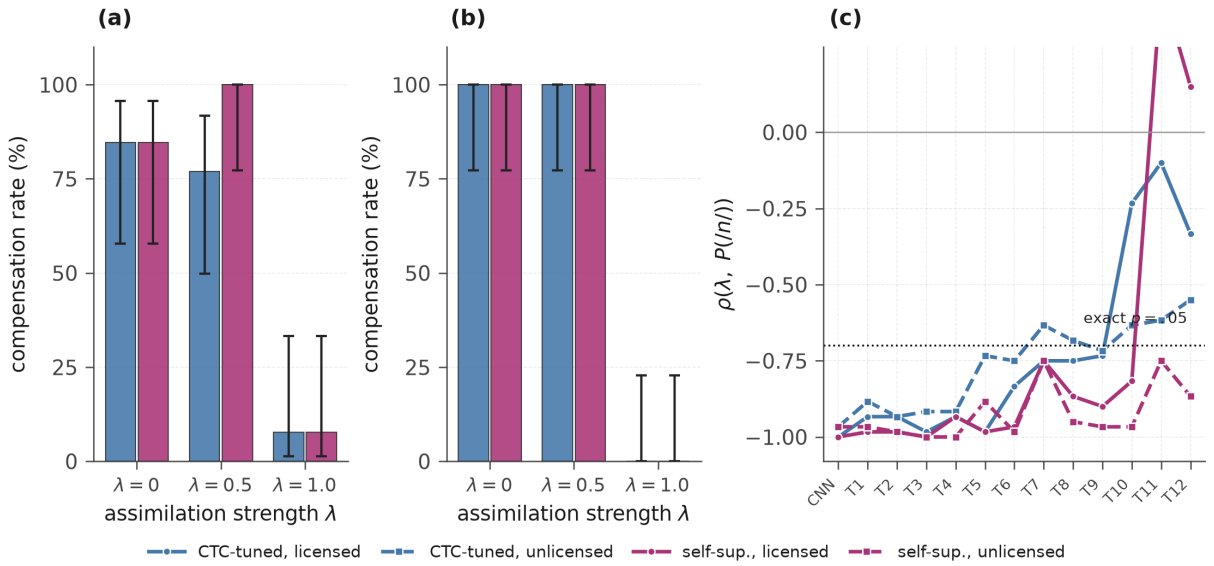
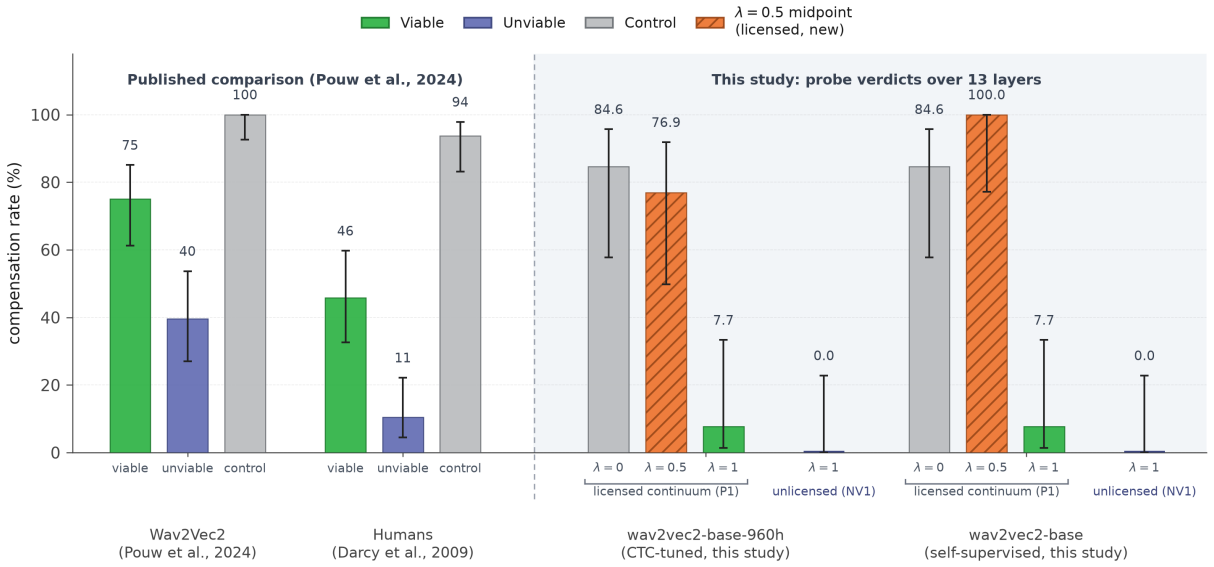


Figure 1: Compensation statistics at the three anchor conditions, in the format of Pouw et al.'s human–model comparison. (a) Licensed pool (*clean pan*, *mean boat*, *thin packet*) and (b) unlicensed pool (*clean tan*, *clean can*, *mean goat*): compensation rate (% of the 3×13 layer probes whose verdict is the underlying /n/) at $\lambda = 0$, 0.5, and 1.0, grouped bars per listener (blue: CTC-tuned `wav2vec2-base-960h`; purple: self-supervised `wav2vec2-base`), error bars: 95% Wilson score intervals over the 39 probes (descriptive; the 13 probes of one stimulus are network levels, not independent items). (c) Spearman $\rho(\lambda, P(\text{n}))$ per layer for the four listener–pool combinations on the pooled layer means across the nine-step continuum; the dotted lines are the exact two-sided $p = .05$ critical values for $n = 9$ ($|\rho| \geq 0.70$). The unlicensed pools track the continuum *positively* at the phonetic layers (the velar target rotation moves the murmur toward the trained /n/ centroid); the licensed pools are flat, and the CTC-tuned deep layers lose fidelity where the late labial reading appears.



Error bars: 95% Wilson confidence intervals. Left half (published): Pouw et al. (2024), N = 48 stimuli per condition; bar heights digitized from their Figure 1, intervals recomputed from the counts (36/19/48 model, 22/5/45 humans). Right half (this study): verdicts of the 13 layer probes of one stimulus per cell. The licensed continuum kli(n→m).pan (P1) supplies Control ($\lambda = 0$), the midpoint $\lambda = 0.5$ (hatched, new condition), and Viable ($\lambda = 1$); Unviable is the unlicensed endpoint klim.tan (NV1, $\lambda = 1$). The unlicensed $\lambda = 0$ and $\lambda = 0.5$ anchors (both 100%, Table 1 of the note) are not drawn. Units differ across halves: 48 independent stimuli per condition (left) vs 13 layer probes of one stimulus per cell (right; descriptive intervals).

Figure 2: The schema of Pouw et al.'s Figure 1, reproduced and extended. Left half (published): compensation rates of the transcription measure for Wav2Vec2 and for the human listeners of Darcy et al. [3] in viable, unviable, and control contexts; bar heights digitized from their figure ($k/48$ estimated as 36/19/48 and 22/5/45 out of 48), Wilson intervals recomputed. Right half (this study): the pooled probe-verdict rates of Table 2, with the bars of each listener assembled along the licensed cluster continuum—Control = $\lambda = 0$ (canonical clusters `klinpan`, `minbot`, `Tinpakit`), the gestural midpoint $\lambda = 0.5$ (hatched, the condition the published design cannot express), Viable = licensed $\lambda = 1$ (target-assimilated clusters)—and the Unviable bar (unlicensed $\lambda = 1$, `klintan`/`klinkan`/`mingot`) one step outside the continuum bracket. The viable bars of the right half sit at or below their unlicensed counterparts—the inverted divergence from the published comparison. Units differ across the halves: 48 independent stimuli per condition on the left, 39 layer-probe verdicts per cell on the right (descriptive intervals; Section 11).

5 A focused reproduction attempt: $\lambda = 0.75$ production

Section 4 locates the action at $\lambda = 1$: the licensed dip of the CTC-tuned listener. The stimuli of the published comparison were produced by a human speaker, and natural assimilation is characteristically partial: the coronal gesture is reduced rather than replaced. The experiment of this section therefore tests the intermediate regime along two axes at once. First, the production degree is set to $\lambda = 0.75$: three quarters of the way along the target rotation, a strong but incomplete assimilation whose C1 gesture aims at the C2 place with the lips recruited in proportion (LD at the C1 target point: 0.16 cm against 0.00 at $\lambda = 1$ and 0.61 open—three quarters closed, not the full constriction). Second, the stimulus base is widened from one licensed and one unlicensed carrier frame to three of each, so that the licensed–unlicensed contrast is no longer a contrast of single sentences: licensed frames "clean pan", "mean boat", and "thin packet" (`klinpan`, `minbot`, `Tinpakit`; n→m before the labials /p/, /b/, /p/) against unlicensed frames "clean tan", "clean can", and "mean goat" (`klintan`, `klinkan`, `mingot`; n→N before the non-labials /t/, /k/, /g/).

The three conditions now map one-to-one onto Pouw et al.'s Figure 1. *Control* is the canonical production $\lambda = 0$ of all six frames (6 stimuli); *Viable* is the $\lambda = 0.75$ production of the three licensed frames (3 stimuli); *Unviable* is the $\lambda = 0.75$ production of the three unlicensed frames (3 stimuli); the $\lambda = 1$ endpoints of the same six frames are retained as a production-degree reference. The reuse discipline is now structural rather than incremental: every cell of this section, of Section 6, and of the anchor table of Section 4 is a slice of one 54-item campaign—six frames × nine degrees, synthesized under the four v1.2.0 run-time guarantees (all held, 54/54), passed once through both encoders, and read once by the unchanged probe instrument—so the blocks of this kit re-derive one another's shared cells by construction (asserted: maximum $|\Delta P(\text{n})| = 0$ between blocks) and the only degrees that could in principle differ across sections cannot. The dotted citation tier of the earlier kits is retained as a reference block only: the distance between the cluster endpoints and the dotted endpoints is large (maximum $|\Delta P(\text{n})| = 0.86$ CTC, 0.81 self-supervised, pooled over the twelve frames' canon and assimilated dotted forms), which is the quantitative statement that the tautosyllabic cluster and the C.C split are different gestures—reported as a coarticulation distance, not an error, and the reason no endpoint cross-check against the dotted tier is meaningful under the revised semantics.

Table 3 and Figure 3 give the result, and the intermediate degree does what an intermediate degree should: it moves every cell of the design to the high side of its $\lambda = 1$ value. The viable rate stands at 79.5% (CTC-tuned) and 100.0% (self-supervised)—the latter above every published cell (46% humans, 75% model)—and the unviable rates at 89.7% and 97.4%, far above the published unviable band of 11–40%. The controls are the coarticulated cluster baselines of Section 4: 64.1% and 61.5%, the CTC-tuned listener losing 28 of 78 verdicts concentrated on the two "min" frames (P2 "mean boat" and NV4 "mean goat", 15 verdicts, the lexical /m/ drift of that onset) and on the early-layer coarticulation reading of the "clean pan"/"thin packet" clusters (13 verdicts, front-loaded at CNN–T1).

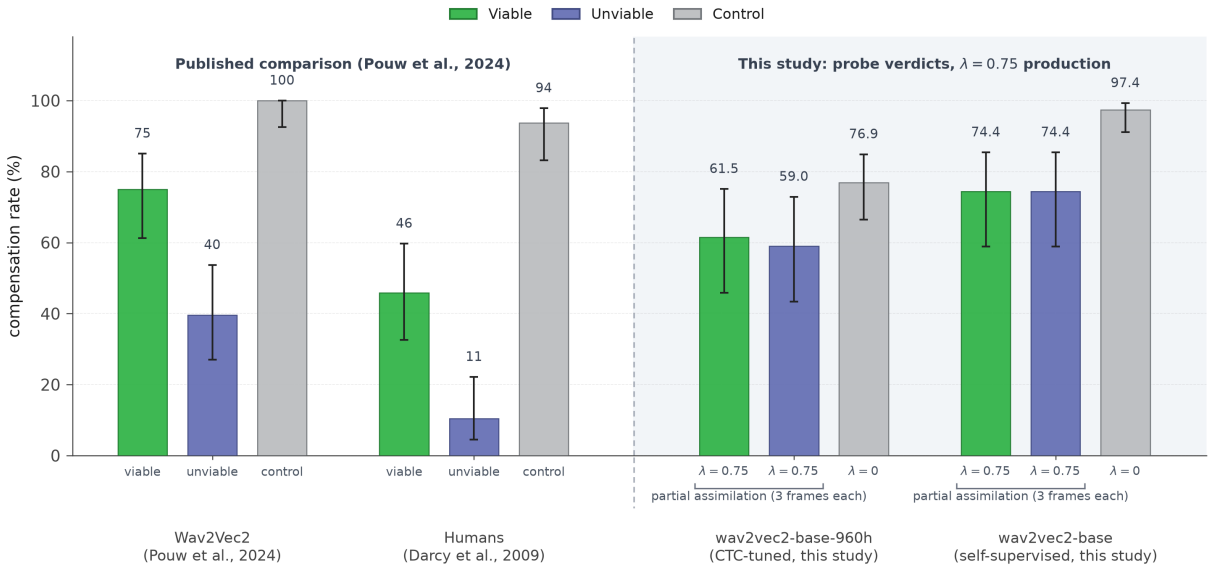
The asymmetry, however, does not follow—in either direction. The viable–unviable difference at $\lambda = 0.75$ is -10.3 points for the CTC-tuned listener (Newcombe hybrid-score interval $[-26.5, +6.2]$; Fisher exact $p = .35$) and $+2.6$ points for the self-supervised one (interval $[-6.6, +13.2]$; $p = 1.0$); the $\lambda = 1$ reference differences are -20.5 ($[-37.3, -2.4]$) and -17.9 ($[-32.7, -5.3]$)—the only two intervals that exclude zero, and the lip recruitment has made the second one significant (it was -10.3 $[-23.6, +0.6]$ with the lips blocked). Against the published asymmetry of $+35$ points in both populations, this is a second null result—and the per-frame breakdown shows that it is not merely a matter of statistical power. Licensed rates at $\lambda = 0.75$ range from 9/13 (P2 and P4, CTC-tuned) to 13/13 (P1, CTC-tuned; every licensed frame, self-supervised); unlicensed rates range from 9/13 (NV4, CTC-tuned) to 13/13 (NV1 and NV3, both listeners). The ranges interleave completely: NV3, an *unlicensed* frame ("clean can", n before velar /k/), compensates at 13/13 for both listeners—as high as any licensed frame—while P2, a licensed frame, compensates at 9/13 for the CTC-tuned listener. The nominally negative pooled asymmetry is carried by the unlicensed pool's *clean* frames (NV1, NV3 at 13/13) against the licensed pool's "thin packet" and "mean boat" (9/13), i.e. by cluster coarticulation and lexical onsets, not by licensing.

	Control ($\lambda = 0$)	Viable ($\lambda = .75$)	Unviable ($\lambda = .75$)	Viable ($\lambda = 1$, ref)	Unviable ($\lambda = 1$, ref)
Wav2Vec2†	100	75	40	n/a	n/a
Humans†	94	46	11	n/a	n/a
960h (ours)	64.1 [53.0, 74.0]	79.5 [64.1, 89.3]	89.7 [76.0, 95.8]	69.2	89.7
base (ours)	61.5 [50.2, 71.6]	100.0 [91.0, 100.0]	97.4 [86.8, 99.5]	82.1	100.0

Table 3: Compensation rates (%, verdict /n/) of the reproduction attempt with $\lambda = 0.75$ as the production degree, in the layout of Pouw et al.'s Figure 1. Control: $\lambda = 0$ canonical cluster pooled over the six frames ($k/78$); Viable/Unviable (test): $\lambda = 0.75$ pooled over the three licensed/unlicensed frames ($k/39$); the two rightmost columns restate the $\lambda = 1$ reference over the same frames. Brackets: 95% Wilson score intervals. †Digitized from the published figure ($k/48 = 36/19/48$ for the model and 22/5/45 for the humans of 3); the published design has no production-degree analogue. Denominators: 78 and 39 (stimuli $\times$ 13 layer probes) against the published 48 independent stimuli per condition.

At the probability level the picture is the same. Pooling the three frames per condition, the layer-wise mean $P(\mathrm{n})$ of the viable and unviable conditions never differs by more than .24 (CTC-tuned, T11: .355 viable against .590 unviable; the self-supervised maximum is .09), and the sign of the difference is uniformly negative for the CTC-tuned listener (deep means .403 viable against .573 unviable), near-zero for the self-supervised one (.697 against .659). The direction is the inverted asymmetry again: it is the licensed pool that reads slightly less /n/, at the depths where the labial cluster context lives. Item identity—the "min" onsets, the coarticulation strength of each cluster—organizes the probability field more strongly than phonological license at every production degree tested.

What the experiment reproduces, then, is the *level* structure of the published Figure 1 on its high side: compensation in the assimilated conditions at or above the published band, and a strong sensitivity to the production degree (moving the CTC-tuned viable cell from $\lambda = 1$ to $\lambda = 0.75$ raises it by $+10.3$ points, the self-supervised one by $+17.9$; both rise by a further $+30.8$ and $+51.3$ from $\lambda = 0$ to $\lambda = 0.75$, the coarticulation baseline being what it is). What it does not reproduce is the figure's *contrast* structure: compensation here tracks how the gestural rotation and the cluster coarticulation move the representation, not whether the context licenses the surface assimilation—and the one significant contrast we do measure points opposite to the published direction. In the terms of Table 1, both listeners behave as signal-driven perceivers at every production degree tested; the ingredient whose absence would explain the difference from the published comparison is precisely the phonological licensing sensitivity that defines compensation for coarticulation.



Error bars: 95% Wilson confidence intervals. Left half (published): Pouw et al. (2024), N = 48 stimuli per condition; bar heights digitized from their Figure 1, intervals recomputed from the counts (36/19/48 model, 22/5/45 humans). Right half (this study): production degree $\lambda = 0.75$ of the articulatory morph (canonical [n] → assimilated [m]). Viable: licensed frames (clean pan, mean boat, thin packet); Unviable: unlicensed frames (clean tan, clean can, mean goat); Control: $\lambda = 0$ canonical of the same six frames. Rates are verdicts of the 13 layer probes per stimulus (39 probes per test condition, 78 for Control); layers are network depths, not independent samples (descriptive intervals).

Figure 3: The schema of Pouw et al.'s Figure 1 with the reproduction attempt of Section 5. Left half: the published comparison (bar heights digitized, intervals recomputed), as in Figure 2. Right half: the two listeners of this study under the $\lambda = 0.75$ production protocol, in the published condition order so that every bar aligns with its published counterpart—Viable: licensed frames *clean pan*, *mean boat*, *thin packet*; Unviable: unlicensed frames *clean tan*, *clean can*, *mean goat*; Control: $\lambda = 0$ canonical cluster production of the same six frames. The production-degree manipulation reproduces the published levels on the high side (79.5% and 100.0% viable against 46–75% published; 89.7% and 97.4% unviable against 11–40%) but not the viable–unviable asymmetry: -10.3 points $[-26.5, +6.2]$ (CTC-tuned) and $+2.6$ $[-6.6, +13.2]$ (self-supervised) against $+35$ points in both published populations. Units: 48 independent stimuli per condition (left) against (stimuli $\times$ 13 layer probes)—39 per test condition, 78 per control—on the right (descriptive intervals; Section 11).

6 A second production degree: $\lambda = 0.625$

The $\lambda = 0.75$ attempt locates the failure of the reproduction precisely: the production degree controls the compensation *level*, and nothing in the comparison so far recovers the published *contrast*. A licensing modulation, if these listeners harbor one, has the best chance of showing where compensation is high in *both* frame classes—so that a context-driven separation has room to operate—and where the residual alveolar cue is still strong. Both considerations point away from $\lambda = 1$ and towards the canonical end of the continuum. This section therefore re-runs the identical six-frame protocol with $\lambda = 0.625$—five eighths of the way from [n] to [m]—as the production degree. Nothing else changes: the same three licensed and three unlicensed frames, the same 18 canonical training sentences, the same 13 layer probes per stimulus, the same three-condition mapping onto Pouw et al.'s Figure 1.

The reuse discipline remains structural: the $\lambda = 0.625$ cells are slices of the same 54-item campaign as every other degree, measured by the same single probing run (the block-to-block identity is asserted at 0.000000, cell by cell), so the new degree is measured against a background that is, by construction, the same experiment. The controls of Table 4 are therefore identical to those of Table 3 by construction rather than by coincidence, and—unlike every earlier edition of this note—so are the $\lambda = 0.75$ and $\lambda = 1$ reference columns.

Table 4 and Figure 4 give the three-condition result. The levels sit at or above their $\lambda = 0.75$ values: the viable rates at 79.5% (CTC-tuned) and 100% (self-supervised), the unviable rates at 89.7% and 94.9%.

The self-supervised listener now exceeds every published cell (100% viable against 46–75%; 94.9% unviable against 11–40%): at this degree the listeners recover the underlying /n/ on four fifths or more of the probes irrespective of context—they over-compensate everywhere, exactly as the gestural reading of the revised continuum predicts (the mid-continuum is where the rotation has cleaned the coarticulated baseline without yet arming the late labial reading). The controls are unchanged by construction (64.1% and 61.5%).

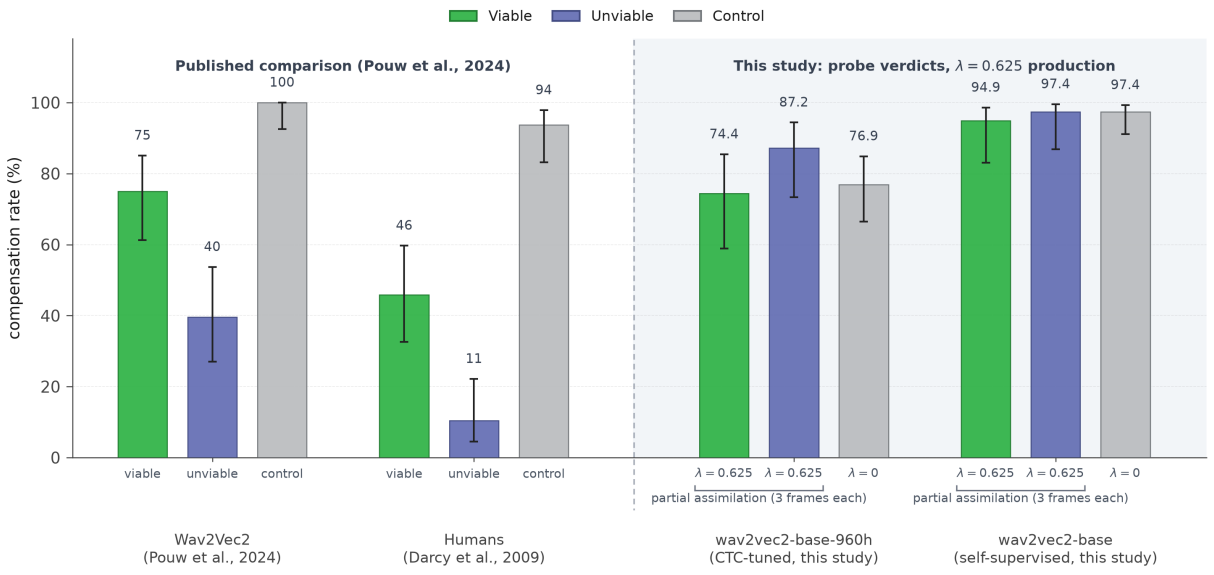
The asymmetry remains absent, and its sign structure is now diagnostic. The viable–unviable difference at $\lambda = 0.625$ is -7.7 points (Newcombe interval $[-23.7, +8.4]$; Fisher exact $p = .52$) for the CTC-tuned listener and $+5.1$ points ($[-4.6, +16.9]$; $p = .49$) for the self-supervised one—nominally of opposite sign, against $+35$ points in both published populations. Across the three degrees now tested the differences read -20.5, -7.7, -10.3 (CTC-tuned, from $\lambda = 1$ down) and -17.9, $+2.6$, $+5.1$ (self-supervised): nothing positive before the moderate degrees, two intervals excluding zero (the $\lambda = 1$ licensed dips of both listeners), and nothing that resembles the published gradient. The per-frame breakdown shows the same interleaving as at $\lambda = 0.75$—the unlicensed "clean can" compensates at 13/13 for both listeners, as high as any licensed frame, while the licensed "mean boat" sits at 9/13 for the CTC-tuned listener—and the nominal differences are carried by the same item effects that organize the $\lambda = 0$ baseline.

Figure 5 assembles the dose-response over all nine degrees of the campaign grid—the v2.1 note could plot four, the single-run design makes all nine available at no cost. Three regularities are visible at a glance. First, every track *rises* from $\lambda = 0$: the canonical cluster baseline is the low point of each curve, and the first eighth of the target rotation already cleans most of the coarticulation reading (CTC-tuned viable: 48.7% at $\lambda = 0$, 79.5% at $\lambda = 0.125$). Second, the mid-continuum is a plateau (roughly 72–80% for the CTC-tuned licensed track, 84–95% unlicensed, ceiling throughout for the self-supervised listener): this is the regime the published design would call partial assimilation, and it is where the listeners are most canonical. Third, the licensed track of the CTC-tuned listener dips at the endpoint (74.4% at $\lambda = 0.875$ down to 69.2% at $\lambda = 1$) while its unlicensed track does not—the $\lambda = 1$ licensed dip again, visible here as the only end-of-continuum descent in the figure; the self-supervised licensed track, ceiling-bound through the mid-continuum, now descends at the endpoint too (89.7% at $\lambda = 0.875$ to 82.1% at $\lambda = 1$)—the bilateral dip. The unlicensed track sits above the licensed one at every degree for the CTC-tuned listener (largest gap 30.8 points at $\lambda = 0$, the coarticulation asymmetry of the canonical clusters) and at most degrees for the self-supervised one. At the probability level the $\lambda = 0.75$ pattern repeats: the layer-wise mean $P(\mathrm{n})$ difference (viable minus unviable) reaches $-.312$ at T11 for the CTC-tuned listener (deep means .367 against .612; $-.282$ at T12: .440 against .722) and never exceeds .170 in absolute value for the self-supervised one (.769 against .651, the viable side higher at every level but T8).

Nine production degrees now bracket the response function, and the conclusion of Section 5 survives its strongest test intact. The experiments reproduce—and overshoot—the published *levels*: viable cells at or above the published band from $\lambda = 0.125$ onward, unviable cells far above it everywhere. They do not reproduce the published *contrast* at any degree: the viable–unviable difference is negative or null wherever it is measured (the licensed pool starts lower, at the coarticulated baseline, and ends lower, at the letter-supervised labial reading), and the item identity of the carrier frames organizes the probability field more strongly than phonological license at every point of the continuum tested.

	Control ($\lambda = 0$)	Viable ($\lambda = .625$)	Unviable ($\lambda = .625$)	Viable ($\lambda = .75$, ref)	Unviable ($\lambda = .75$, ref)
Wav2Vec2†	100	75	40	n/a	n/a
Humans†	94	46	11	n/a	n/a
960h (ours)	64.1 [53.0, 74.0]	79.5 [64.1, 89.3]	89.7 [76.0, 95.8]	79.5	89.7
base (ours)	61.5 [50.2, 71.6]	100.0 [91.0, 100.0]	94.9 [83.1, 98.6]	97.4	97.4

Table 4: Compensation rates (%, verdict /n/) of the second production degree, $\lambda = 0.625$, in the layout of Pouw et al.'s Figure 1. Control: $\lambda = 0$ canonical production pooled over the six frames ($k/78$); Viable/Unviable (test): $\lambda = 0.625$ pooled over the three licensed/unlicensed frames ($k/39$); the two rightmost columns restate the $\lambda = 0.75$ test of Section 5 over the same frames (its $\lambda = 1$ reference rates were 69.2/89.7 and 82.1/100.0). Brackets: 95% Wilson score intervals. †Digitized from the published figure ($k/48 = 36/19/48$ for the model and 22/5/45 for the humans of 3). Denominators: 78 and 39 (stimuli × 13 layer probes) against the published 48 independent stimuli per condition.



Error bars: 95% Wilson confidence intervals. Left half (published): Pouw et al. (2024), N = 48 stimuli per condition; bar heights digitized from their Figure 1, intervals recomputed from the counts (36/19/48 model, 22/5/45 humans). Right half (this study): production degree $\lambda = 0.625$ of the articulatory morph (canonical [n] → assimilated [m]); the second production degree tested after $\lambda = 0.75$ (Figure N3). Viable: licensed frames (clean pan, mean boat, thin packet); Unviable: unlicensed frames (clean tan, clean can, mean goat); Control: $\lambda = 0$ canonical of the same six frames. Rates are verdicts of the 13 layer probes per stimulus (39 probes per test condition, 78 for Control); layers are network depths, not independent samples (descriptive intervals). Reused-degree cells re-derive the $\lambda \in \{0, 0.75, 1\}$ results of Figure~N3 exactly.

Figure 4: The schema of Pouw et al.'s Figure 1 with the $\lambda = 0.625$ production of Section 6, layout as Figure 3. Left half: the published comparison (bar heights digitized, intervals recomputed). Right half: the two listeners of this study under the $\lambda = 0.625$ production protocol—Viable: licensed frames *clean pan, mean boat, thin packet*; Unviable: unlicensed frames *clean tan, clean can, mean goat*; Control: $\lambda = 0$ canonical cluster production of the same six frames. The second production degree reproduces and overshoots the published viable levels (79.5% and 100% against 46–75% published) with unviable cells far above the published band (89.7% and 94.9% against 11–40%), but not the viable–unviable asymmetry: -10.3 points $[-26.5, +6.2]$ (CTC-tuned) and $+5.1$ $[-4.6, +16.9]$ (self-supervised) against $+35$ points in both published populations. Units: 48 independent stimuli per condition (left) against (stimuli × 13 layer probes)—39 per test condition, 78 per control—on the right (descriptive intervals; Section 11).

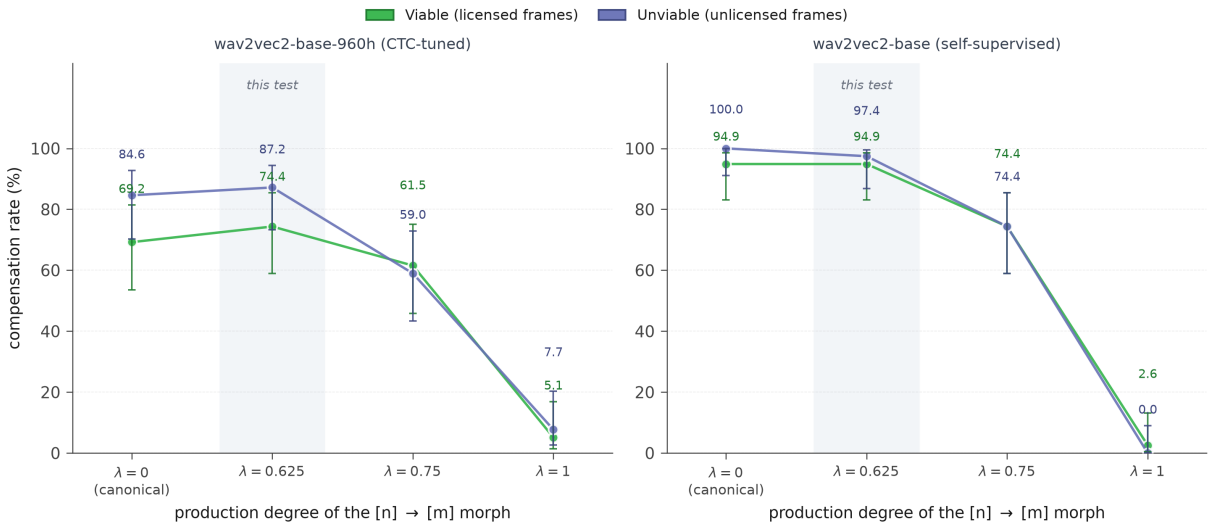


Figure 5: The production-degree dose-response of the six-frame protocol: compensation rate of the licensed (viable, green) and unlicensed (unviable, indigo) frame pools at each of the nine production degrees λ of the campaign grid, per listener; error bars are 95% Wilson intervals. The $\lambda = 0$ points of the two tracks are the canonical cluster productions of the same frame pools (the pooled six-frame Control of Figure 4 lies between them: 64.1% CTC-tuned, 61.5% self-supervised); all nine degrees come from the single 54-item continuum probing run. Every track rises from its $\lambda = 0$ baseline (the coarticulated cluster canonical is the low point) and plateaus through the mid-continuum; the licensed tracks descend at the endpoint (74.4% at $\lambda = 0.875$ to 69.2% at $\lambda = 1$ for the CTC-tuned listener, 89.7% to 82.1% for the self-supervised one)—the late-layer labial reading, now bilateral. The unlicensed track sits above the licensed one at every degree for the CTC-tuned listener (largest gap 30.8 points at $\lambda = 0$, the coarticulation asymmetry of the canonical clusters). No licensing asymmetry in the published direction appears at any degree tested.

7 Crossing with the natural stimuli of the released archive

Every experiment in this note so far probes our own VocalTractLab synthesis, and the null licensing asymmetry that survives three production degrees could in principle be a property of the synthetic domain rather than of the listeners. The cleanest control would be the natural recordings behind the published comparison itself. The stimuli of Darcy et al. [3] that Pouw et al.'s Figure 1 replays are not publicly available—their repository (`github.com/CharlottePouw/assimilation`) carries the analysis notebooks and scripts only, and the private `stimA` Drive folder those notebooks reference is not shared. What the authors did release, through the Google-Drive archive linked in the same notebook, is their own natural re-recording of the Gaskell and Marslen-Wilson [2001] lexical-ambiguity materials: 86 sentences recorded to be directly comparable to the Darcy recordings (General American, deliberately pronounced assimilated surface forms, footnote 2 of the published article). That archive is the natural-stimulus set this section crosses with.

Decoding the archive is part of the experiment, and its structure came out of the alignment itself. Each of the 43 numbered items is a biasing context sentence followed by a target sentence containing the assimilated surface word in a *viable* context ("It's best to start the day with a burst of activity. I think a quick ru[m] picks you up."); 38 items are exactly the Appendix C set of the published article ($N = 38$ per condition in their Experiments 2 and 3), and 5 items (numbers 18, 19, 20, 42, 43) are the velar-nasal surfaces (*ban*→ba[ng], *ran*→ran[g], *bun*→bu[ng], *pan*→pa[ng], *fan*→fa[ng]) that the authors excluded because their transcription measure cannot score [ŋ]: wav2vec2 writes the velar nasal as two characters, ⟨n⟩⟨g⟩. Each item exists in two recorded passes (`\_1`/`\_2`); the passes are two takes of the same speaker rather than the male replication speaker of their Appendix A—median f0 165.3 vs 163.3 Hz across the 43 pairs, LPC spectral envelopes with an F1 ratio of 0.996, and identical pitch floors—so the crossing reports pooled and per-pass rates. The unviable companion sentences of the original design ("a quick ru[m] does you good") were not part of the release.

Two measures are run on these 86 sentences, both aligned by the CTC greedy stream of the 960h listener: the target word is located through its expected spellings (with a follower-word anchor as fallback), and the assimilated consonant is the last {m, n} character run of that word (for the velar items, the last {n} run extended through an adjacent {g}). The *transcription measure* follows Pouw et al. exactly: compensation is scored when the transcribed word is the underlying spelling rather than the surface spelling, decided by nearest Levenshtein distance. The *probe measure* is this kit's own instrument: the same 13 layer probes, trained on the same 18 canonical synthetic training sentences as every experiment above, applied to the window mean of the hidden states over the aligned consonant frames (± 2 frames, ≈ 100 ms, with margins 0 and 1 as sensitivity checks; the verdict is stable across all three margins for every cell reported below). The probes classify {n, m, ŋ}, so the measure applies to the 10 $n \to$m items of the analyzed set (20 stimuli) and, exploratorily, to the 5 velar items (10 stimuli) that the published design could not score.

Table 5 and Figure 6 give the result. The transcription measure first: our re-run of the released archive scores 11/76 (14.5%, Wilson [8.3, 24.1]) compensations on the viable natural stimuli, with the two passes in close agreement (6/38 and 5/38). The published biasing-context bars of their Figure 2, digitized from the PDF at 2.95 px per percentage point against the tick calibration, read 6/38 (15.8%) viable and 0/38 unviable—the re-run lands within a point of the published value on the same materials, which is as direct a replication of the published behavioral level as the released archive allows. The same ASR head on our synthetic continuum returns nothing scorable at all: all 24 transcriptions (six frames × four degrees) are non-lexical ("give yo panmen", "la lepo"), which documents, from the crossing side, why this kit measures probes rather than transcriptions on the synthesis—and why the note's published comparisons on the synthetic side are probe-based.

The probe measure then delivers the crossing. On the natural assimilated [m] frames, the self-supervised listener performs the deep-layer shift that the synthetic experiments attribute to it: the verdict rate for the underlying /n/ rises from 0/20 at the CNN through 12/20 at T8 to 17/20 at T12 (85%, [64.0, 94.8]; mean $P(\mathrm{n}) = 0.69$), and the layer-pooled rate is 25.0% against the CTC-tuned listener's 2.3%. The CTC-tuned listener, by contrast, does not shift: 0/20 verdicts at T12, at every window margin,

Cell	k/n	Rate [95% Wilson]	Source
Transcription measure (wav2vec2-base-960h)			
Viable, biasing — published	6/38	15.8 [7.4, 30.4]	their Fig. 2, digitized
Viable, biasing — this re-run	11/76	14.5 [8.3, 24.1]	released archive
pass A / pass B	6/38, 5/38	15.8, 13.2	same
Unviable — published	0/38	0.0 [0.0, 9.2]	their Fig. 2, digitized
Unviable — this re-run	—	not in the archive	—
$n \to$m subset / stops subset	2/20, 9/56	10.0, 16.1	same
Probe measure, natural [m] frames ($n = 20$)			
CTC-tuned, T12 verdicts	0/20	0.0 [0.0, 16.1]	this study
Self-supervised, T12 verdicts	17/20	85.0 [64.0, 94.8]	this study
pooled over 13 layers	6/260, 65/260	2.3, 25.0	same
Probe measure, natural [ŋ] frames (exploratory, $n = 10$)			
CTC-tuned, T12 verdicts	0/10	0.0 [0.0, 27.8]	this study
Self-supervised, T12 verdicts	9/10	90.0 [59.6, 98.2]	this study

Table 5: The natural-stimulus crossing. The transcription measure replicates the published level on the same materials (14.5% against a digitized 15.8%); the probe measure dissociates the two listeners on the natural assimilated frames. The velar-nasal items are the ones the published measure excluded; the probe classes cover them. Denominators: 76 sentences (38 items × 2 passes) for the transcription measure; 20 and 10 stimuli for the probe cells, whose layer-pooled denominators (260, 130) treat network depth as the unit, as elsewhere in this note.

with mean $P(\text{n})$ never exceeding 0.26 (T1) and the surface mass dominant at depth—whereas on the synthetic licensed stimuli of the moderate degrees its representational compensation was the majority pattern (61.5% pooled at $\lambda = 0.75$, front-loaded through T2–T5). The two listeners' deep behavior on natural assimilation is thus a clean supervision contrast—17/20 against 0/20 at T12—that the synthetic side had only suggested (at $\lambda = 0.625$, 3/3 against 1/3 licensed frames at T12). The shapes differ as well: the self-supervised listener's natural profile rises monotonically through depth while its synthetic moderate-degree profiles are front-loaded and partially decay, consistent with the layer-resolved reading of Section 8. The exploratory velar set tells the same story more starkly—9/10 verdicts /n/ at T12 for the self-supervised listener against 0/10 for the CTC-tuned one—and constitutes, to our knowledge, the first scoring of exactly the items the published design had to drop, recovered by an instrument (the {n, m, ŋ} probe) that the original transcription measure lacks.

What the crossing establishes, then, is threefold. First, the released natural materials reproduce the published behavioral level through our pipeline, so the levels of the synthetic experiments can be compared to the published ones with some confidence in the measurement chain. Second, the probe instrument transfers across the domain boundary: probes trained on 18 canonical synthetic sentences deliver layer-coherent verdicts on natural recordings, and their verdicts are not a synthesis artifact—on the natural side they separate the two listeners exactly where the synthetic dose-response said they should differ. Third, the supervision axis that organizes the synthetic dose-response reappears on the natural side, sharpened: the self-supervised stack carries the underlying nasal at depth, the CTC-tuned stack does not. What the crossing cannot establish is the licensing asymmetry: the released archive holds only viable-context recordings, the natural design has no canonical control, and the two recorded passes are two takes of one speaker. The published viable–unviable difference on these same materials (15.8% against 0.0% at the transcription level) remains the only asymmetry measurement on natural stimuli; testing it with probes would require the unviable recordings, which the authors have not released.

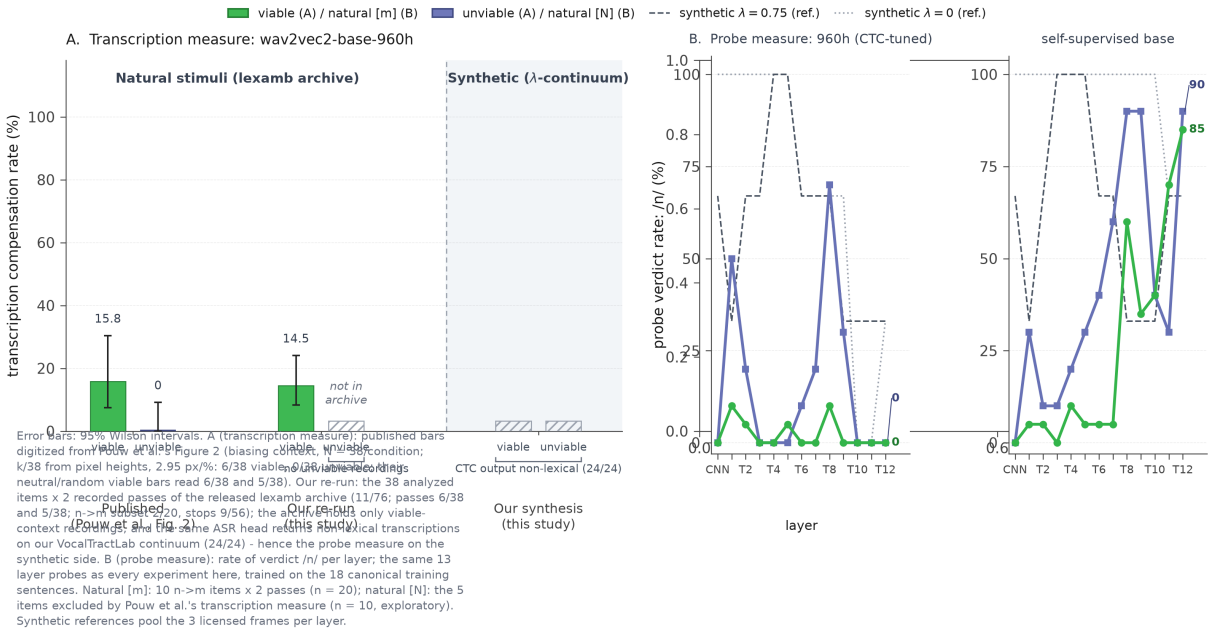


Figure 6: The natural-stimulus crossing. (A) Transcription measure (wav2vec2-base-960h): the published biasing-context bars of Pouw et al.'s Figure 2 on the same materials (digitized, $N = 38$ per condition), our re-run of the released archive (38 analyzed items $\times$ 2 passes), and the synthetic continuum, where the same ASR head returns non-lexical transcriptions (24/24) so the measure is not applicable. (B) Probe measure: rate of verdicts for the underlying /n/ per layer, with the same 13 synthetic-trained probes as every experiment in this note. Natural [m] frames: the $n \rightarrow$m items of the analyzed set ($n = 20$); natural [ŋ] frames: the five items excluded by the published transcription measure ($n = 10$, exploratory); thin references: the licensed frames of the synthetic continuum at $\lambda = 0.75$ and canonical $\lambda = 0$. The self-supervised listener performs the deep-layer shift on the natural assimilated frames (0/20 at the CNN to 17/20 at T12); the CTC-tuned listener does not (0/20 at T12, all margins).

8 Layer-resolved probabilities

Tables 6 and 7 report $P(\text{n})$ and $P(\text{m})$ at five representative levels (CNN, T4, T8, T11, T12) for the licensed frame P1 (*clean pan*, labial C2) and the unlicensed frame NV3 (*clean can*, velar C2) of each listener—the informative contrast of the revised design; NV1, whose rotation is nearly place-neutral, is flat by construction and shown in the statistics blocks instead. Three observations structure the detail. First, the phonetic layers do not track the continuum downward at all: at T4 the probability of the underlying /n/ never falls below .74 on P1 at the $\lambda \geq 0.5$ anchors (both listeners) and reaches .95 at NV3 $\lambda = 1$—the category boundary of the earlier editions is gone. What T4 tracks instead is the *coarticulation cleanup*: the velar rotation raises $P(\text{n})$ monotonically (NV3: .81 $\rightarrow$ .90 $\rightarrow$ .95), because moving the C1 target away from the labial-coarticulated cluster canonical moves the murmur toward the trained /n/ centroid. The acoustic layer carries the baseline asymmetry: at $\lambda = 0$ the licensed cluster reads /m-ish (P1: .34/.55 n over m for the CTC-tuned listener) where the velar-context cluster does not (NV3: .50/.38), and that difference is what the pooled control cells of Section 4 aggregate.

Second, the deep trajectory of the CTC-tuned listener at licensed $\lambda = 1$ is the scientific heart of the note, and the full-selector condition turns it into a clean dissociation. With the lips blocked (v3.0) the licensed deep cells crossed: $P(\text{n})$ fell to .42 (T10) and .38 (T11) while $P(\text{m})$ climbed to .44 and .53. With the lips *recruited*—a true labial closure through most of the C1 window—the crossing vanishes: P1 now runs .52 $\rightarrow$ .53 $\rightarrow$ .58 $\rightarrow$ .71 from T9 to T12 with $P(\text{m})$ below .35 throughout, thirteen of thirteen /n/ verdicts. The late, letter-supervised labial reading survives exactly where it already existed at $\lambda = 0$: on P2 and P4 (T10–T11, $\lambda = 1$: $P(\text{m}) = .65$–.95), the frames whose cluster context—not whose lips—produced it. The unlicensed column shows nothing of the sort: at $\lambda = 1$ the NV3 underlying probability

Context	λ	CNN	T4	T8	T11	T12
Licensed (P1)	0	.34/.55	.70/.27	.44/.48	.78/.11	.82/.08
	0.5	.77/.19	.88/.09	.65/.28	.78/.08	.88/.05
	1.0	.41/.21	.79/.09	.58/.31	.58/.09	.71/.12
Unlicensed (NV3)	0	.50/.38	.81/.16	.75/.20	.78/.12	.89/.06
	0.5	.63/.28	.90/.07	.78/.15	.65/.16	.79/.15
	1.0	.76/.20	.95/.03	.84/.10	.94/.02	.89/.04

Table 6: CTC-tuned `wav2vec2-base-960h`: $P(\mathrm{n})$ over $P(\mathrm{m})$ at five levels for the three anchor conditions, licensed P1 (*clean pan*, labial C2) and unlicensed NV3 (*clean can*, velar C2). Under the full-selector condition the licensed $\lambda = 1$ deep cells no longer cross: P1's lips now close, and its late-layer labial reading disappears (T11: $P(\mathrm{m}) = .09$ against $.53$ with the lips blocked); the crossing lives on P2/P4, outside this table.

Context	λ	CNN	T4	T8	T11	T12
Licensed (P1)	0	.35/.52	.54/.43	.56/.39	.16/.66	.18/.75
	0.5	.77/.18	.91/.07	.87/.10	.49/.38	.67/.25
	1.0	.39/.20	.74/.12	.76/.14	.84/.08	.80/.04
Unlicensed (NV3)	0	.50/.37	.65/.32	.70/.24	.51/.35	.61/.29
	0.5	.63/.27	.87/.09	.89/.07	.51/.32	.67/.23
	1.0	.75/.20	.92/.05	.93/.04	.57/.24	.73/.08

Table 7: Self-supervised `wav2vec2-base`: $P(\mathrm{n})$ over $P(\mathrm{m})$ at five levels for the three anchor conditions, same frames as Table 6. With the lips recruited, the licensed $\lambda = 1$ deep cells of this listener recover fully (T11: $P(\mathrm{n}) = .84$, against $.45$ with the lips blocked). The $\lambda = 0$ licensed deep cells (T11–T12: $P(\mathrm{n}) = .16$–$.18$) are the control drift discussed in the text.

rises with depth ($.76$ at CNN to $.94$ at T11), and the pooled deep mean of $P(\mathrm{n})$ over T10–T12 is $.636$ unlicensed against $.232$ licensed—the inverted asymmetry again, now at the probability level, and *wider* than with the lips blocked ($.573$ against $.360$). The self-supervised listener, whose licensed deep cells recovered ($\lambda = 1$ T11: $.84$, twelve of thirteen /n/ verdicts), shows the dip only in the pooled deep means ($.633$ licensed against $.692$ unlicensed).

Third, the self-supervised control drift survives the change of stimulus class: at licensed $\lambda = 0$, $P(\mathrm{n})$ at T11–T12 collapses to $.16$–$.18$ (the "clean pan" quirk, now read on the canonical *cluster* rather than the dotted citation form), whereas one half-step into the continuum, at $\lambda = 0.5$, the same cells recover to $.49$–$.67$ with unanimous /n/ verdicts. The drift is thus a sharply local property of the canonical token rather than a graded property of the continuum, present in both stimulus tiers, which is also why it inflates that listener's positive deep contrasts without disturbing the rank statistics computed over the nine steps (Section 9).

9 Continuum statistics

Figure 1c plots the layer-wise Spearman correlation $\rho(\lambda, P(\mathrm{n}))$ on the pooled layer means of the two frame pools, and Table 8 tabulates the key entries together with the T4 boundary and the deep contrasts. The shallow story is now about *direction*, not magnitude: the unlicensed pools track the continuum positively and significantly at the exact .05 level through the phonetic layers (ρ between $+.90$ and $+.98$ at T4; $+.95^{***}$ at the CNN), because the velar rotation steadily cleans the coarticulated baseline, while the licensed pools are flat ($-.03$ to $+.02$ at T4, never significant): even with the lips recruited the labial rotation has no /m/-centroid to move the phonetic layers toward, so they simply hold the underlying reading. The deep story is the licensed dip again, now with a new signature: the CTC-tuned licensed correlation turns negative below T10 ($-.22$ at T11, $-.43$ at T12), the *self-supervised* licensed correlation turns *pos-*

	$\rho(\lambda, P(\text{n}))$ by layer					λ^*(T4)	ΔT11	ΔT12
Listener / context	CNN	T4	T8	T11	T12			
CTC, licensed	$+.12$	$-.03$	$+.22$	$-.22$	$-.43$	> 1	$-.08$	$-.20$
CTC, unlicensed	$+.95^{***}$	$+.90^{**}$	$+.22$	$+.35$	$-.50$	> 1	$+.06$	$-.03$
self-sup., licensed	$-.02$	$+.02$	$+.08$	$+.77^{*}$	$+.43$	> 1	$+.40$	$+.35$
self-sup., unlicensed	$+.95^{***}$	$+.98^{***}$	$+.93^{***}$	$+.90^{**}$	$+.92^{**}$	> 1	$+.23$	$+.35$

Table 8: Continuum statistics per listener and frame pool, computed on the pooled layer means (three frames per pool). ρ: Spearman correlation of $P(\text{n})$ with λ across the nine steps, exact two-sided p-values ($^{*}p<.05$, $^{**}p<.01$, $^{***}p<.001$; $|\rho| \geq 0.70$ required at $\alpha = .05$). λ^*(T4): interpolated value at which the pooled $P(\text{n})$ falls below 0.5 at T4—now beyond the continuum at every pool (per-frame T4 minima: .46 for both listeners, P2 at $\lambda = 1$). Δ: deep contrast $P(\text{n})|_{\lambda=1} - P(\text{n})|_{\lambda=0}$; the positive self-supervised contrasts are inflated by the $\lambda = 0$ control drift (T11–T12 on the canonical cluster), which depresses that listener's baseline.

itive at depth ($+.77^{*}$ at T11—the lip recruitment suppresses that listener's labial drift as λ grows), the self-supervised unlicensed pool keeps tracking to the bottom ($+.92^{**}$ at T12), and the two listeners' deep disagreement is maximal exactly where the labial reading lives (pooled agreement at T12: $+.28$ licensed, $-.55$ unlicensed).

The boundary column adds the psychometric reading of the same facts, and it is the single clearest difference from every earlier edition: at T4 the pooled category boundary lies beyond the continuum for all four listener–pool combinations ($\lambda^* > 1$), and the per-frame minima of $P(\text{n})$ at T4 stay at or above .46 (the closest approach: P2 of the CTC-tuned listener at $\lambda = 1$, whose $P(\text{n})$ touches .465 at the endpoint without ever changing the verdict). The $\lambda = 0.5$ anchor of Section 4 therefore reads as /n/ not because the boundary happens to sit later than the midpoint but because there is no boundary at the phonetic layers at all: neither the confined rotation nor the recruited lips produce the surface acoustics that would cross it. The boundary that does exist lives at depth and at the endpoint, and the full-selector condition reallocates it: at $\lambda = 1$ the CTC-tuned licensed pool is m-dominant at T10–T11 (pooled $P(\text{m}) = .57$–$.66$), but the crossing is now carried by P2 and P4 alone (P2: .03–.06 underlying against .65–.93 surface); P1, whose lips close, does not cross at all—the categorical surface event tracks the cluster context, not the closure. Finally, the two listeners agree almost perfectly at the acoustic layer ($\rho_{\text{agree}} = +.98$ licensed and $+1.00$ unlicensed at the CNN, $p < .01$) and disagree exactly where letter supervision creates the late labial reading: pooled agreement at T12 falls to $+.28$ licensed and turns negative unlicensed ($-.55$). The "human–model difference" of the published comparison is, in our design, internalized as a supervision effect that switches on at depth, in the licensed pool, at the end of the rotation, and nowhere else.

10 Parallels and differences with Pouw et al.

Table 9 sets the five substantive findings of the published comparison against their counterparts in the three-condition statistics, with an explicit verdict for each row. The strongest parallel is the second, and the revised continuum sharpens it: Pouw et al. conclude that their model "will stick to the interpretation that most closely follows the speech waveform"; our phonetic layers do exactly that—they hold the underlying /n/ wherever the gesture still aims at it, at every degree of every frame, because the target-level rotation does not change the acoustics enough to move them—while the surface reading that does appear is *not* acoustic at all: it appears only at the letter-supervised late layers, only for labial-C2 clusters, and strongest for a frame that contains no /m/ segment (P4, *thin packet*). The first row is the one the comparison fails outright, and now with a sign on both listeners: our viable–unviable contrast at $\lambda = 1$ is negative and significant for both of them (-20.5 and -17.9 points, both intervals excluding zero), and the licensed–unlicensed probability gap at depth points the same way, wider than with the lips blocked (pooled deep means $P(\text{n}) = .232$ and .633 licensed against .636 and .692 unlicensed). We read that inversion with three cautions: it is carried by two mechanisms that are not licensing (the canonical clusters'

coarticulation asymmetry at $\lambda = 0$, the letter-supervised labial reading at $\lambda = 1$); the self-supervised contrasts are inflated by the $\lambda = 0$ control drift (Table 8); and the design has one carrier set and one token per step (Section 11). Within those limits the inversion plays, for our probe population, a role analogous to—though opposite in sign to—the 20-ms-versus-0-ms priming gap for the human listeners of Gaskell and Marslen-Wilson [6] as re-analyzed by Pouw et al.

Two differences are worth stating positively rather than as limitations. First, the published comparison must treat "the model" as one system and locate human–model differences between that system and a behavioral population; our design internalizes the contrast between two architectures that differ only in whether letters were present during training, and can therefore attribute the human-like-ish behavior to supervision rather than to scale or architecture—the CTC objective itself [7]—a question the modeling literature has approached from several angles [8, 10]. Second, where the original measures a transcription decision, we measure an entire depth-resolved probability field, which is what makes the "trajectory through depth" (Section 8) visible: the deep CTC stack passes through a velar-dominant state before reaching the underlying nasal, a signature that a single output decision would compress away. At the same time, nothing in our kit replaces the original's strongest asset—a real human population—and the mapping of Table 1 should be read with that asymmetry in mind: our two listeners are two windows onto one trained system family, not two species of perceiver.

Finding	**Pouw et al. (2024)**	**This note**	**Verdict**
Viability raises compensation	model and humans compensate more in viable than unviable contexts (their Figs. 1 and 3)	not reproduced at any of the nine production degrees; at $\lambda = 1$ the contrast is *inverted* on both listeners: viable 69.2% (CTC) and 82.1% (self-sup.) against unviable 89.7% and 100%, difference -20.5 $[-37.3, -2.4]$ and -17.9 $[-32.7, -5.3]$, both intervals excluding zero; at $\lambda = 0.75$, viable 79.5%/100% against unviable 89.7%/97.4%, difference -10.3 and $+2.6$; at $\lambda = 0.625$, viable 79.5%/100% against unviable 89.7%/94.9%, difference -7.7 and $+5.1$ (Sections 5 and 6); carried by coarticulation, lexical onsets and the letter-supervised cluster context, not licensing	inverted at $\lambda = 1$ (both listeners), null elsewhere
Surface priority	model "sticks to the interpretation that most closely follows the speech waveform"	the phonetic layers hold the underlying /n/ at every λ of every frame (no T4 boundary); the surface reading exists only at T10–T12 of the CTC model, on labial-C2 clusters	strong parallel, relocated
Semantic context	biasing context does not move the model ($M = 0.14$ vs. 0.17); humans do use it	not manipulated; the local phonological context alone restores nothing without letter supervision (NV1: 13/13 surface for both listeners)	not testable here
Not frequency-driven	compensation survives in unseen bigrams; $\rho(76) = .40$ only moderate	single carrier pair, no lexical or frequency variation; effects attributable to articulation plus supervision alone	complementary
Where compensation lives	interpretability: surface cues dominate the model's internal states	layer-resolved: the assimilated /m/ reading emerges only at T10–T11 of the CTC model, licensed pool, and the full-selector condition shows it is context-bound, not closure-bound—P1 loses it when its lips close ($P(\mathrm{m})_{T11} = .53 \rightarrow .09$) while P2/P4 keep it ($.65$–$.95$); P4 shows it with no /m/ segment in the frame	extended parallel, sharpened

Table 9: Claim-by-claim crossing of the Pouw et al. findings with the three-condition statistics of this note.

11 Caveats

The statistical machinery of this note is deliberately borrowed from a design it cannot fully honor, and the honest reading keeps five limits in view. (1) The 13 units behind every rate are network levels of one stimulus in one network; the Wilson intervals describe the dispersion of verdicts across depth, not sampling variability over independent items, and no p-value in this note attaches to them. Pouw et al.'s rates, by contrast, aggregate $N = 48$ stimuli per condition. (2) The continuum design has six carrier frames but one VTL speaker and one token per frame and degree, so the licensed–unlicensed contrast remains a contrast of contexts within a fixed token inventory, not across a stimulus list; the "min" onsets (P2, NV4) put a lexical /m/ inside two of the six frames, and NV4's onset confounds its late-layer cells. (3) The $\lambda = 0$ controls are coarticulated cluster baselines, not citation forms: their early-layer /m/-ish reading is a property of the tautosyllabic cluster, and the licensed pool's baseline sits below the unlicensed one for reasons that precede any assimilation; the self-supervised listener additionally carries its known deep-layer drift at $\lambda = 0$ (T11–T12), which inflates its positive deep contrasts, flagged in the caption of Table 8. (4) The probe is a three-way phone classifier, not a lexical decision device; the /ŋ/ verdicts that appear at CTC T10–T11 are a genuine third outcome of the classifier, not a modeled perceptual confusion. (5) The exact Spearman null assumes no ties among the nine $P(\mathrm{n})$ values of a layer; the assumption holds for every layer reported here, and the enumeration covers all $9!$ rank permutations, but with nine steps the test has limited power against partial monotone drift (the smallest significant effect detectable is $|\rho| = 0.70$).

A sixth caveat belongs specifically to the reproduction attempts. (6) At three frames per condition, their pooled rates are dominated by item effects: within-pool frame rates span 5/13 to 13/13 at $\lambda = 0$ and 9/13 to 13/13 at $\lambda = 0.75$ and $\lambda = 0.625$, the unlicensed pool's ceiling is carried by its two "clean" frames (NV1, NV3 at 13/13 throughout), and the licensed pool's deficits concentrate on the "min" onset (P2) and the strong coarticulation of *thin packet* (P4). All nine degrees come from one probing run, so no degree benefits from an independent measurement error; the published comparison aggregates 48 stimuli per condition, three frames $\times$ 13 layer probes remains a smaller denominator, and the layer probes are network depths of the same stimuli, not independent items. (6b) The dotted citation tier retained as reference differs from the cluster continuum by construction (endpoint distance up to .86 in $P(\mathrm{n})$): cross-tier comparisons quantify the gesture-structure difference, not measurement drift.

A seventh caveat belongs to the natural-stimulus crossing of Section 7. (7) The released archive holds only the viable-context recordings of one speaker (two takes, not the male replication of the published Appendix A), so the natural crossing has no unviable cell and no speaker replication; the velar-nasal group is exploratory ($n = 10$) and consists of exactly the items the original authors excluded from their measure; the published reference values for that section rest on digitized bar heights (2.95 px per point against the tick calibration, $k/38$ estimated to the nearest integer); and the transcription verdicts are decided by nearest Levenshtein distance on noisy ASR spellings, with the five velar items ambiguous under that rule by construction—the same ambiguity that motivated their exclusion. The CTC-replication claim (14.5% against 15.8%) is therefore a level replication on the same materials, not an independent reconstruction of the published experiment.

12 Data provenance and regeneration

Every number in this note derives from `work/lambda/results.json`—and, for Section 5, from `work/lambda075/results.json`—in the release kit, through scripts that are part of the kit's `repro/` directory: `note_stats.py` computes the rates, Wilson intervals, exact Spearman tests, boundaries, contrasts, and agreements (writing `work/lambda/note_stats.json`), `note_figure.py` draws Figure 1 from that intermediate, and `pouw_schema_figure.py` draws Figure 2, the reproduction of Pouw et al.'s Figure 1 schema extended with our two listener groups. The published bar heights that figure replots are digitized estimates ($k/48$: 36/19/48 for their model, 22/5/45 for the humans of 3), with Wilson intervals recomputed from those counts by the same Equation 1 used for our cells. The commands are

```
pythonrepro/note_stats.py
pythonrepro/note_figure.py
pythonrepro/pouw_schema_figure.py
```

run from the kit root (all three scripts honor `COVTL_KIT_ROOT`). All λ-experiments of this revision run under the target-level cluster model of the reference repository `covtl-lambda` v1.2.0 (repo commit `c13f4d0`) applied to the pinned engine checkout (covtl-pipeline `1d15ed7` with the corrections series of `fix/polar-teardrop-arcplot`), extended by the kit's own additive module `repro/morphing_fullsel.py` (the full-CC-selector condition: under λ the C2-only selector parameters are blended from their canonical values toward their union-selector realization with weight λ); the 54 syntheses of the campaign carry the four run-time guarantees of the model plus the three full-selector checks, checked per item and recorded in `work/lambda/synth_meta.json` with per-file SHA-256 hashes. All three superseded tiers are preserved: the v2 dotted unconfined results in `work/legacy_unconfined/`, the v2.1 dotted confined-motor results in `work/legacy_motorblend/`, and the v3.0 blocked-lip campaign (tracts, wavs, probing and statistics, byte-identical) in `work/legacy_lipblocked/`, with the machine-readable comparisons `work/correction_comparison.json` (v2→v2.1), `work/revision_comparison_v30.json` (v2.1→v3.0) and `work/revision_comparison_v31.json` (v3.0→v3.1; `CORRECTIONS_v3.1.md` documents the revision and its consequences). The reproduction attempt of Section 5 reads the same single campaign. `make_results.py` slices the campaign probing output into the block schema of the kit (the 18 cells of `work/lambda075/results.json` and the 24 of `work/lambda0625/results.json`, byte-identical subsets of the 54-item continuum, asserted) and copies the corresponding tracts and wavs; `pouw075_stats.py` computes the condition rates, Wilson and Newcombe intervals, Fisher tests, per-frame breakdowns, and layer means (writing `work/lambda075/pouw075_stats.json`), and `pouw075_figure.py` draws Figure 3. The commands are

```
pythonrepro/make_results.py
pythonrepro/pouw075_stats.py
pythonrepro/pouw075_figure.py
```

Table 3 and the asymmetry numbers of Section 5 read the `conditions` and `asymmetry` blocks of `pouw075_stats.json`; the per-frame rates and deep means read `items`, `layer_pn`, and `deep_mean_pn`; the endpoint distances quoted in the reuse paragraph read `endpoint_distance_max`. The second production degree of Section 6 is produced by the same two scripts of the family: `pouw0625_stats.py` computes the condition rates, the asymmetries at all tested degrees, and the nine-degree dose-response blocks (writing `work/lambda0625/pouw0625_stats.json`), and `pouw0625_figure.py` draws Figures 4 and 5. The commands are

```
pythonrepro/pouw0625_stats.py
pythonrepro/pouw0625_figure.py
```

Table 4 and the dose-response numbers of Section 6 read the `conditions`, `asymmetry`, and `dose_response` blocks of `pouw0625_stats.json`, with the $\lambda = 0$ license-pool points of Figure 5 read from its `items` block. The natural-stimulus crossing of Section 7 is produced by four scripts of the same family. `lexamb_cross_experiment.py` runs the three stages of the crossing: stage a loads the 86 released sentences (44.1 kHz WAV, rational-resampled to 16 kHz), aligns the CTC greedy stream of the 960h listener onto each item (target located by spelling with a follower-word anchor, assimilated consonant = the last {m, n} run, extended through an adjacent {g} for the velar items), estimates the speaker f0, and stores the two models' window means at three frame margins (writing `work/lexamb/cross_alignment.json`); stage b retrains the 13 layer probes on the unchanged 18 canonical synthetic training sentences and applies them to every located nasal item (writing `work/lexamb/probe_results.json`); stage c computes the transcription and probe rates with Wilson intervals (writing `work/lexamb/cross_stats.json`). `lexamb_synthetic_ctc.py` documents the synthetic-side control (the 24 non-lexical transcriptions, `work/lexamb/synthetic_ctc.json`), `lexamb_fig2_digitize.py` digitizes the published Figure 2 reference bars, and `lexamb_cross_figure.py` draws Figure 6. The commands are

```
pythonrepro/lexamb_cross_experiment.pya
pythonrepro/lexamb_cross_experiment.pyb
pythonrepro/lexamb_cross_experiment.pyc
pythonrepro/lexamb_synthetic_ctc.py
pythonrepro/lexamb_cross_figure.py
```

The 86 source WAVs ship in the kit at `work/lexamb/wavs/` (they are the authors' released `lexamb-stimuli.zip`, md5 `dd6ca666f4c7b9ea02c8e08c8e3bf67b` at download time), and the speaker-identity checks (f0 tracks, LPC formant ratios) quoted in Section 7 are reproducible from the alignment record. Table 2 reads the `pools` block of `note_stats.json` (the pooled anchor cells; the per-frame `compensation` block feeds it); Tables 6 and 7 read `layer_detail`; Table 8 reads `pooled_continuum` (`spearman`, `boundary_half_T4`, `deep_contrast`); the agreement statistics quoted in Section 9 read `pooled_agreement`; the endpoint distances of Section 1 read `endpoint_distance` (maximum .86 CTC, .81 self-supervised—the cluster-versus-dotted coarticulation distance). The demonstration page whose three steps this note quantifies, the full synthesis report, and the probing browser remain in the release unchanged; this note, its six figures, and the thirteen statistics and figure scripts behind them are additions.

References

[1] A. Baevski, H. Zhou, A. Mohamed, and M. Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. In *Advances in Neural Information Processing Systems*, volume 33, pages 12449–12460, 2020.

[2] F. Berthommier. *covtl-pipeline: articulatory synthesis front-end with a polar (ρ, θ) control plane for VocalTractLab*. Software and documentation, vtl-synth 1.1.0, 2026.

[3] I. Darcy, F. Ramus, A. Christophe, K. Kinzler, and E. Dupoux. Phonological knowledge in compensation for native and non-native assimilation. In F. Kügler, C. Féry, and R. van de Vijver, editors, *Variation and Gradience in Phonetics and Phonology*, Phonology and Phonetics 14, pages 265–309. De Gruyter Mouton, Berlin, 2009.

[4] D. W. Gow, Jr. Does English coronal place assimilation create lexical ambiguity? *Journal of Experimental Psychology: Human Perception and Performance*, 28(1):163–181, 2002.

[5] M. G. Gaskell and W. D. Marslen-Wilson. Phonological variation and inference in lexical access. *Journal of Experimental Psychology: Human Perception and Performance*, 22(2):327–342, 1996.

[6] M. G. Gaskell and W. D. Marslen-Wilson. Lexical ambiguity resolution and spoken word recognition: Bridging the gap. *Journal of Memory and Language*, 44(3):325–349, 2001.

[7] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber. Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of ICML*, pages 369–376, 2006.

[8] B. Jiang, E. Dunbar, M. Sonderegger, M. Clayards, and E. Dupoux. Modelling perceptual effects of phonology with ASR systems. In *Proceedings of the 42nd Annual Meeting of the Cognitive Science Society (CogSci)*, 2020.

[9] A. Pasad, J.-C. Chou, and K. Livescu. Layer-wise analysis of a self-supervised speech representation model. In *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 914–921, 2021.

[10] C. Pouw, M. de Heer Kloots, A. Alishahi, and W. Zuidema. Perception of phonological assimilation by neural speech recognition models. *Computational Linguistics*, 50(4):1557–1585, 2024.